

Muhammad Hassan Arshad

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building production AI/ML systems end-to-end, from trained predictive models to deployed Generative AI pipelines. Designed and trained Scikit-learn classification and regression models served via a dedicated Python/FastAPI microservice, and integrated LLMs (LLaMA 3.3 70B via Groq) and diffusion models (FLUX.1, Stable Diffusion XL) into production mobile applications with robust prompt engineering, structured output validation, and fault-tolerant API handling. Comfortable across the full ML lifecycle: synthetic data generation, feature engineering, model training/evaluation, and microservice deployment, as well as applied NLP (intent extraction, fuzzy matching, recommendation systems). Builds fast without compromising on system design by combining strong engineering fundamentals with AI-augmented development.

TECHNICAL SKILLS

Languages: Python, JavaScript / TypeScript

Machine Learning: Scikit-learn (Random Forest, Gradient Boosting), Pandas, NumPy, Model Training & Evaluation, Synthetic Data Generation, Feature Engineering

Generative AI / LLMs: LLaMA 3.3 70B (Groq API), Hugging Face Inference API, Stable Diffusion XL, FLUX.1, Pollinations API, Prompt Engineering, Structured Output Validation (Zod)

NLP & Search: Intent Extraction, Fuzzy String Matching (RapidFuzz, Levenshtein Distance, Jaccard Index), MongoDB Text Search & TF-IDF Relevance Scoring

ML Deployment & Microservices: FastAPI, Uvicorn, Pydantic, Node.js, Express.js, RESTful API Design, Microservices Architecture

Databases & Caching: MongoDB Atlas (Mongoose, Aggregation Pipelines, Text Indexes), Redis

Security & Reliability: JWT Authentication, bcrypt, Helmet, Rate Limiting, Retry/Backoff Logic, Error Handling

Tools: Git, GitHub, Postman, Cloudinary, Multer, node-cron

Core Competency: End-to-End ML Lifecycle · Generative AI Integration · AI-Augmented Development

PROJECTS

Easy Car — AI/ML Automotive Diagnostics Platform

Stack: Python · FastAPI · Scikit-learn · Pandas · NumPy · Groq (LLaMA 3.3 70B) · Hugging Face · Node.js · MongoDB Atlas

- Architected a microservices system decoupling a Python/FastAPI ML inference service from the main Node.js/Express backend to serve real-time vehicle diagnostics predictions.
- Trained and deployed 3 Scikit-learn models: a Random Forest Classifier predicting real-time vehicle risk level (Low/Medium/High) from engine temperature, RPM, and DTCs; a Gradient Boosting Classifier predicting the most likely failing component; and a Random Forest Regressor producing a continuous 0–100 vehicle health score.
- Built a synthetic data generation pipeline using NumPy statistical distributions (Normal, Poisson, Beta) to produce 5,000 realistic automotive telematics records, then engineered an automated training pipeline with an 80/20 train-test split, evaluated via accuracy score and mean absolute error.
- Deployed all models behind a production-ready FastAPI server with Pydantic request validation, exposing a /predict REST endpoint consumed by the main backend.
- Integrated LLaMA 3.3 70B via the Groq API to power a conversational diagnostics assistant that translates raw OBD-II fault codes into plain-language explanations and repair recommendations, using dynamic prompt engineering grounded in live vehicle state data.
- Built a Generative AI vehicle customization pipeline (Hugging Face FLUX.1-schnell, Pollinations Flux) that generates photorealistic 3D modification previews from natural-language prompts, with exponential backoff for model cold-starts and rate-limiting to protect costly generation endpoints.

Penny Wise — AI-Powered Price Discovery Engine

Stack: Python · FastAPI · RapidFuzz · Groq (LLaMA 3.3 70B) · Redis · Node.js · MongoDB Atlas

- Built a conversational shopping assistant using LLaMA 3.3 70B via Groq for low-latency inference, performing intent extraction from natural-language queries into structured JSON (validated with Zod) — automatically extracting category, price range, brand, and keywords.
- Maintained multi-turn conversational memory using Redis for stateful, context-aware chatbot interactions.

- Engineered a cross-platform product-matching pipeline combining fuzzy string matching (RapidFuzz, Levenshtein distance) in a Python microservice with Jaccard Index token-matching in Node.js, to deduplicate and link identical products scraped across multiple e-commerce platforms.
- Built a personalized recommendation engine via scheduled batch jobs (node-cron), computing per-user category affinity scores from search and view history, with results pre-cached for instant ($O(1)$) read latency on the dashboard.
- Implemented full-text relevance search across an 80,000+ product catalog using MongoDB Text Indexes with TF-IDF-style textScore ranking.
- Designed a decoupled scraper microservice (Python, FastAPI, BeautifulSoup4, httpx, APScheduler) to handle high-IO data ingestion independently from the core API, keeping the platform available during large-scale scraping runs.

ADDITIONAL EXPERIENCE

- Full Stack Development Intern, CodeAlpha (Remote) — Delivered 3 production-grade full-stack web applications (real-time video conferencing platform, social media platform, e-commerce platform) using Node.js, Express, MongoDB, Socket.io, and WebRTC.

EDUCATION

Bachelor of Science in Software Engineering

Expected June 2026

University of Central Punjab — Lahore, Pakistan

- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Software Engineering, Cloud Computing, Cyber Security, System Design & Architecture